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Tecnologías de la información y comunicación

INTE/ISO/IEC 9241-210:2019

Ergonomics of person interaction system - Part 210: User-centered design in the person for systems interactive.

Correspondence: This national standard is identical (IDT) to the International standard ISO 9241-210:2019, "Ergonomics of human-system interaction -- part 210: human-centred design for interactive systems.

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INTERNATIONAL PROLOGUE

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the procedures used to prepare this document and those intended for its subsequent maintenance is described in the ISO/IEC Directives, Part 1. In particular, they should be taken into account count the different approval criteria required for the various types of ISO documents. This document was drafted in accordance with the editorial standards of the ISO/IEC guidelines. Part 2 (verwww.iso.org/directives).

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For an explanation of the voluntary nature of the norms, the meaning of the terms and specific expressions of the ISO related to conformity assessment, as well as information about the ISO's adherence to the principles of the World Trade Organization (WTO) on Technical Barriers to Trade (TBT), seeewww.iso.org/iso/foreword.html

This document was prepared by the ISO Technical Committee ISO/TC 159, Ergonomics, Subcommittee SC 4. Ergonomics of the person-system interaction.

Any comments or questions regarding this document should be directed to the national agency of normalization of the user. A complete list of these organizations can be found in www.iso.org/members.html

This second edition cancels and replaces the first edition (ISO 9241-210:2010), of which constitutes a minor revision. The changes compared to the previous edition are as follows:

Figure 1 has been updated for greater clarity;

In section 7.1, additional information on accessibility has been added;

Editorial changes have been made to align with ISO/IEC Directives, Part 2.

A list of all the parts of the ISO 9241 series can be found on the ISO website.

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NATIONAL PROLOGUE

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The representation of all sectors involved in the Technical Standardization process is guaranteed by the Technical Committees and the Public Consultation period, the latter characterized by the participation of the general public.

This standard has been developed in compliance with the level 1 and level 2 requirements of the Standards. Council of Canada (SCC).

This standard INTE/ISO/IEC 9241-210:2019 was approved by the National Normalization Commission. from INTECO on the date of 2019-12-18.

This standard is subject to permanent updates in order to ensure that it responds to all moment to the current needs and demands.

Below are the companies that collaborated in the study of this standard through from its participation in the Technical Committee CTN 03 SC 02, Web Accessibility.

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0 INTRODUCTION

User-centered design is an approach to the development of interactive systems that has
with the goal of making systems usable and useful, focusing on the users,
your needs and requirements, and applying human/ergonomic factors, and knowledge and
Usability techniques. This approach increases effectiveness and efficiency, improves human well-being.
the satisfaction of users, accessibility and sustainability, and counteracts the
possible adverse effects of use on human health, safety, and performance.

There is a substantial body of human/ergonomic factors and usability knowledge.
about human-centered design. The aim of this document is to make this available
information to assist those responsible for managing the design and redesign processes of
hardware y software a identificar y planificar actividades de diseño efectivas y oportunas centradas
in the person.

The person-centered design approach described in this document complements the
existing system design approaches. It can be incorporated into such diverse approaches as
object-oriented, waterfall development methods and rapid application development.

The principles of human-centered design and the related activities have not changed.
substantially since the ISO 13407 standard was developed and have been validated for ten years
application. This document reflects this by making requirements and recommendations.

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Ergonomics of the human-system interaction - part 210: human-centered design for interactive systems.

1 OBJECT AND SCOPE OF APPLICATION

This document provides requirements and recommendations for the principles and activities of person-centered design throughout the entire life cycle of interactive systems based on computers. It is intended to be used by those who manage the design processes and it deals with the ways in which the hardware and software components of interactive systems interact they can improve the interaction between person-system.

Note. Computer-based interactive systems vary in scale and complexity. Examples include ready-to-use software products, customized office systems, process control systems automated banking systems, websites and applications, and consumer products such as machines vending machines, mobile phones, and digital television. Throughout this document, these systems are referred to generally products, systems, or services, although, for simplicity's sake, sometimes only one is used term.

This document provides an overview of person-centered design activities. Not provides a detailed overview of the methods and techniques required for user-centered design in the person, nor does it address in detail the aspects of health or safety. Although it covers planning and the management of person-centered design does not address all aspects of management projects.

The information contained in this document is intended for those responsible for planning and project management that designs and develops interactive systems. Therefore, it addresses the human technical factors and ergonomic issues only to the extent necessary for these people can understand their relevance and importance in the design process in their set. It also provides a framework for human factors professionals and usability that participates in person-centered design. The detailed issues of factors Human/ergonomic, usability and accessibility are addressed in more detail in a series of standards, including other parts of the ISO 9241 standards (see Annex A) and INTE/ISO 6385, which they establish the general principles of ergonomics.

The requirements and recommendations in this document can benefit all parties. involved in person-centered design and development. Annex B contains a list of control that can be used to support claims in accordance with the present document

2STANDARDS OF REFERENCE

There are no regulatory references in this document.

TERMS AND DEFINITIONS

For the purposes of this document, the following terms and definitions apply.
ISO and IEC maintain terminological databases for use in standardization in the following addresses:

- ISO Online Navigation Platform: available athttps://www.iso.org/obp
- IEC Electropedia: available athttp://www.electropedia.org/

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3.1 accessibility

degree to which products, systems, services, environments, and facilities can be used by the people of a population with the widest range of needs, characteristics and user capabilities (3.1) to achieve the objectives identified in the contexts identified use (3.10)

Note 1 at the entrance: The context of use includes direct use or use supported by assistive technologies.

[Source: ISO 9241-112:2017, 3.15]

3.2 context of use

combination of users, goals and tasks, resources and environment

Note 1 to the entry: The "environment" in a usage context includes the technical, physical, social, and cultural environments. organizational.

[Source: INTE/ISO 9241-11:2018, 3.1.15]

3.3 effectiveness

the accuracy and thoroughness with which people achieve the specified objectives

[Source: INTE/ISO 9241-11:2018, 3.1.12]

3.4 efficiency

the resources used in relation to the results obtained

Note 1: Typical resources include time, human effort, costs, and materials.

[Source: INTE/ISO 9241-11:2018, 3.1.13]

3.5 ergonomics/ human factors

scientific discipline that deals with the understanding of the interactions between human beings and other elements of a system, as well as the profession that applies the theory, the principles, the data and design methods in order to optimize human well-being and the overall performance of the system

Source: INTE/ISO 6385:2016, 2.3

3.6 objective

expected result

[Source: INTE/ISO 9241-11:2018, 3.1.10]

3.7 person-centered design

approach to the design and development of systems aimed at making interactive systems become more usable, focusing on system use, applying human/ergonomic factors, knowledge and usability techniques

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Note 1 at the entry: The term 'human-centered design' is used instead of 'user-centered design'.
"user" to underline that this document also addresses the impacts on a number of stakeholders, not only in those considered typically as users. However, in practice, these terms are often used as synonyms.

Note 2 to the entry: Usable systems can provide a number of benefits, including which are
they include the improvement of productivity, the increase in users' well-being, the avoidance of stress, the increase in accessibility and the reduction of the risk of damage.

3.8 interactive system

combination of hardware and/or software and/or services and/or people with which users interact to achieve specific objectives

Note 1 to the entry: This includes, when applicable, the packaging, the user's documentation,
online and human help, support and training.

Source: INTE/ISO 9241-11: 2018, 3.1.5

3.9 prototype

interactive system representation of all or part of an interactive system which, although limited In some way, it can be used for analysis, design, and evaluation.

Note 1 at the entrance: A prototype can be as simple as a sketch or a static model or as complicated. like an interactive system in full operation with more or less complete functionality.

3.10 satisfaction

to what extent the physical, cognitive, and emotional responses of the user that result the use of a system, product, or service meets the needs and expectations of the person user.

Note 1 at the entrance: Satisfaction includes the extent to which the user's experience results the real use meets the needs and expectations of the user.

Note 2 to the entry: Early use may influence satisfaction with actual use.

Source: INTE/ISO 9241-11:2018, 3.1.14]

3.11 stakeholder

individual or organization that has a right, participation, claim, or interest in a system or in their possession of characteristics that meet their needs and expectations

[Source: ISO/IEC/IEEE 15288:2015, 4.1.44, modified - The example and note 1 have been omitted] entry.

3.12 task

set of activities undertaken to achieve a specific goal

Note 1 at the entrance: These activities can be physical, perceptual, and/or cognitive.

Note 2 at the entry: Although the objectives are independent of the means used to achieve them, the tasks describe particular means to achieve them.

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Source: INTE/ISO 9241-11:2018, 3.1.11

3.13 usability

the extent to which a system, product, or service can be used by specific users to achieve specific objectives with effectiveness, efficiency, and satisfaction in a specific usage context

Note 1 at the entry: The specified users, the objectives, and the context of use refer to the particular combination of users, objectives, and context of use for which usability is being considered.

Note 2 to the entry: The word "usability" is also used as a qualifier to refer to the knowledge, competencies, activities, and design attributes that contribute to usability, such as ["usability experience","usability professional","usability engineering","usability method","evaluation"] usability, usability heuristic.

Source: INTE/ISO 9241-11:2018, 3.1.1

3.14 user

person who interacts with a system, product, or service

Note 1 to the entry: The users of a system, product, or service include the people who they operate the system, the people who make use of the system output and the people who provide support to the system (including the provision of maintenance and training).

Source: INTE/ISO 9241-11:2018, 3.1.7

3.15 user experience

the perceptions and responses of users that result from the use and/or anticipated use of a system, product, or service

Note 1 to the entry: The perceptions and responses of users include emotions, beliefs, preferences, perceptions, comfort, behaviors, and achievements of users before during and after use.

Note 2 at the entry: The experience of the user is a consequence of the brand image, presentation, functionality, system performance, interactive behavior, and capabilities of assistance of a system, product, or service. It is also the result of the internal and physical state of the person user would result from previous experiences, attitudes, skills, abilities, and personality; and from context of use.

Source: INTE/ISO 9241-11:2018, 3.2.3

3.16 user interface

all the components of an interactive system (software or hardware) that provide information and controls for the user to perform specific tasks with the system interactive

[Source: INTE/ISO 9241-110:2006]

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3.17 validation

confirmation, through the provision of objective evidence, that the requirements have been fulfilled for a specific intended use or application

Note 1 at the entrance: Validation is the set of activities that ensure and gain trust that a the system is capable of achieving the intended use, goals, and objectives (that is, meeting the requirements of the stakeholders) in the planned operational environment.

[Source: INTE/ISO 9000:2015, 3.8.13, modified - Note 1 of the entry has been replaced and notes 2 and 3 from the entry have been removed.

3.18 verification

confirmation, through the provision of objective evidence, that the requirements have been met specified

Note 1 at the entry: Verification is a set of activities that compares a system or element of system with the required characteristics. This may include, but is not limited to, specific requirements, description of the design and the system itself.

Source: INTE/ISO 9000:2015, 3.8.12, modified - Note 1 of the entry has been replaced and notes 2 and 3 of the entry have been removed.

JUSTIFICATION OF THE ADOPTION OF THE CENTERED DESIGN IN THE PERSONA

The use of a person-centered design and development approach has important economic and social benefits for users, employers, and providers.
Highly usable systems and products tend to be more successful both technically and commercially. In some areas, such as consumer goods, buyers will pay a first for well-designed products and systems. The costs of support and technical assistance are they are reduced when users can understand and use the products without additional help.
In most countries, employers and providers have a legal obligation to protect users of health risks, and safety and methods focused on the people can reduce these risks (for example, musculoskeletal risks). The systems designed with person-centered methods improve overall quality, for example, by:

a) increase the productivity of users and the operational efficiency of the organizations;

b) to be easier to understand and use, thus reducing training and support costs;

c) increase usability (effectiveness, efficiency, and satisfaction)

d) increase accessibility (for people from a population with the widest range of needs, characteristics, and capabilities of the users

e) improve the user experience;

f) reduce discomfort and stress;

g) providing a competitive advantage, for example, by improving brand image;

h) contribute to sustainability goals.

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The person-centered approach can lead to higher person-centered quality. (usability, accessibility, user experience, avoiding harm derived from use), such as defined in the standard ISO 9241-220.

The full benefits of person-centered design can be determined by taking into account the total life cycle costs of the product, system, or service, including the conception, the design, implementation, support, use, maintenance, and finally, disposal. The adopting a person-centered design approach contributes to other aspects of design systems, for example, improving the identification and definition of functional requirements. The adoption of a person-centered design approach also increases the likelihood of complete the project successfully, on time and within budget. The use of appropriate methods Person-centered approaches can reduce the risk that the product does not meet the requirements of the stakeholders or being rejected by its users.

Table 1 illustrates examples of results from person-centered design activities.

Table 1 - Examples of results from person-centered design activities

Activities	Design results person-centered	Examples of information contained in the products
Understand and specify the usage context	Context Description of - use	Group Profiles of users Scenarios as they are People
Specify the description requirements user	of identified needs users needs of the person of the user Specification of the requirements of the person - user	User needs derivatives required design guide
Produce design solutions to meet these requirements user-system	Specification of Usage interaction Interface Specification - High de usuario Interface of user implemented	scenarios Low-fidelity prototypes Fidelity Prototypes
Evaluate the designs against the requirements	Evaluation Results Test report of Usability testing results in accordance Follow-up results - Survey report of in the long term	 Field report users
Note. You can find more detailed information about each output in ISO/IEC TR 25060.		

5 PRINCIPLES OF PERSON-CENTERED DESIGN

5.1 Generalities

This document provides a framework for person-centered design. It does not assume any specific design process, nor describes all the different activities necessary to ensure an effective design of the systems. It is complementary to design methodologies existing and provides a person-centered perspective that can be integrated into different design and development processes in a manner appropriate to the particular context. All the person-centered design activities identified in Chapter 7 are applicable (to a greater the least extent) at any stage of system development.

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Whatever the design process and the assignment of responsibilities and roles adopted, a person-centered approach should follow the principles listed below (and that are described in sections 5.2 to 5.7):

a) the design is based on an explicit understanding of the users, the tasks and the environments (see section 5.2);

b) users participate in the entire design and development (see section 5.3);

c) the design is driven and refined through a person-centered evaluation user (see section 5.4);

d) the process is iterative (see section 5.5);

e) the design refers to the entire experience of the user (see section 5.6);

f) the design team includes multidisciplinary skills and perspectives (see section 5.7).

5.2 The design is based on an explicit understanding of the users, the tasks and the environments

Products, systems, and services should be designed with the people who use them in mind. they will use, as well as other interested groups, including all those who may be affected (directly or indirectly) through its use. Therefore, all user groups should be identified. and relevant stakeholders. The construction of systems based on an understanding inappropriate or incomplete in meeting the needs of users is one of the main sources of failures in systems.

The extent to which products are usable and accessible depends on the context, that is, the specified user personas, who have specific objectives and perform specific tasks in a specific environment (see INTE/ISO 9241-11). For example, the type of interface that provides a good user experience for a young person downloading music on a phone can be completely inappropriate to access corporate data on a digital personal assistant (PDA, for its acronym in English). The characteristics of user individuals, the tasks, and the environment. they are called context of use. Section 7.2 provides guidance on how to collect información pertinente. El contexto de uso es una fuente importante de información para establecer the requirements (see section 7.3) and an essential contribution to the design process.

5.3 Users involved in the design and development

Involving users in the design and development provides a valuable source of knowledge about the context of use, the tasks and the way in which users can work with the future product, system, or service. The participation of the users should being active, whether by participating in the design, serving as a source of relevant data, or evaluating solutions. The people involved should have abilities, characteristics, and experience that reflect the range of users for whom the system is being designed. The nature and the frequency of this participation may vary throughout the design and development, depending on the type of the project. The effectiveness of user participation increases as it increases the interaction between developers and users.
When custom systems are developed, the intended users and the tasks performed they can be directly linked to the development process. The organization that acquires the system you have the opportunity to have a direct influence on the design as it emerges, and those those who are really going to work with the future system can participate in the evaluation of the proposed solutions.

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This participation can also increase the acceptance and commitment of users.

When generic or consumer products are developed, the user population becomes dispersed and the products can target groups of users with particular characteristics. It remains It is important that users or appropriate representatives participate in the development. so that the requirements of the users and the relevant tasks can be identified for the group or groups of users intended, in order to include them in the system specification and provide feedback through testing the proposed design solutions.

5.4 Design driven and refined by a user-centered evaluation

User feedback is a critical source of information in design. person-centered. Evaluating designs with users and improving them based on your comments provides an effective means to minimize the risk that a the system does not meet the needs of users or the organization (including the requirements that are hidden or difficult to specify explicitly). This evaluation allows that the preliminary design solutions be tested in 'real world' scenarios, and that the results feed back into increasingly refined solutions. The evaluation focused on the user persona should also have a place as part of the final acceptance of the product for confirm that the requirements have been met. User feedback during Operational use identifies long-term problems and provides information for future design.

Note. The term 'user-centered' is used here to emphasize that this evaluation is carried out from the perspective of the user.

National note: For Costa Rica, the translation for user-centred will be person-centred. user and user as a user person to ensure greater inclusion in the use of language.

5.5 Iterative process

The most appropriate design for an interactive system cannot normally be achieved without iteration.

Note 1. In this context, iteration means repeating a sequence of steps until the result is achieved. desired.

Note 2. In development methods that consist of mini-development cycles, the activities centered on The person can be iterated for individual parts of the system and again at the macro level throughout. the product, system or service.

The iteration should be used to progressively eliminate uncertainty during development. of interactive systems. Iteration implies that the descriptions, specifications, and prototypes they are reviewed and refined when new information is obtained to minimize the risk that the the system under development does not meet the user's requirements.

The complexity of human-computer interaction means that it is impossible to specify complete and exactly every detail of every aspect of the interaction at the beginning of the development. Many of the needs and expectations of users and other stakeholders that will influence the design of the interaction only arise during the course of development, as the designers refine their understanding of users and their tasks, and as they users are in better conditions to express their needs in response to the possible solutions.

La iteración de las soluciones propuestas, incorporando la retroalimentación desde la perspectiva de the user provides a means to mitigate the risk.

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EXAMPLE 1 Feedback from the user's perspective is used to
update the intended usage context, review the requirements and refine the design solutions
proposals.

EXAMPLE 2 The specification of the requirements is iteratively refined using scenarios,
early mockups and prototypes, to obtain feedback from users on whether
These incorporate the requirements of the correct and complete user.

The interaction between the person-centered design aspects and other aspects of it.
It can also lead to the need for iteration, for example, to take manufacturing into account.
of a product, the impact on the production environment or changes in the market.

5.6 Design taking into account the user's experience

The user experience is the result of the presentation, functionality, performance of
system, interactive behavior and assistance capabilities of an interactive system, both of
hardware as well as software. It is also a consequence of the person's previous experiences.
user, their attitudes, skills, habits, and personality. There is a misconception that the
usability refers solely to making products easy to use. However, the
The concept of usability used in ISO 9241 is broader and, when interpreted from the
perspective of the personal objectives of users, may include the type of aspects
perceptual and emotional aspects typically associated with the user experience, as well as
issues such as job satisfaction and the elimination of monotony.

The design for the user experience involves considering, when appropriate, the
organizational impacts, user documentation, online help, support and
the maintenance (including help desks and customer contact points), the
training, long-term use, and product packaging (including the 'experience of
product ready to use"). The user's experience should also be taken into account.
with previous systems or other systems and issues such as branding and advertising. The necessity
Considering these different factors and their interdependencies has implications for the plan of
project (See Chapter 6).

The strengths, limitations, preferences, and expectations of users should be taken into account.
regarding the time to specify what activities the users carry out and what
functions carried out by technology.

Note 1. In critical security and mission-critical systems, it may be more important to ensure effectiveness or efficiency of the system that meets the user's preferences.

The design decisions related to this assignment of functions determine to what extent
a specific job, task, function, or responsibility should be automated or assigned to
human performance. Decisions are based on many factors. These include capabilities.
and relative limitations of humans against technology in terms of reliability,
speed, accuracy, strength, responsiveness, financial cost, the importance of the
successful or timely completion of tasks, the safety and satisfaction of users
both in the short term, for example, as comfort and pleasure, and in the long term, for example, as
health, well-being, and job satisfaction). Making such decisions solely based on the functions
that technology is capable of performing and then simply assigning the remaining functions of
a system for users is likely to result in an ineffective design. The assignment of
functions are described in more detail in section 7.4.2.2.

In general, representative users should participate in these decisions.

"Representative" in this context means that it appropriately corresponds to the target population of
end users.

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The resulting human activities should form a set of tasks that are meaningful.
as a whole for the users. This is particularly important for systems
custom organizational arrangements in which the use of the system supports the main elements
of the tasks (jobs) of the users. For more information, see the ISO 9241-2 standard and the
ISO 10075 series.

5.7 Design team with multidisciplinary skills and perspectives

User-centered design teams don't have to be large, but the team should
be diverse enough to collaborate in the design and implementation of decisions
of compensation at the right moments. The following areas of skills and points of
Views may be necessary in the design and development team:

a) human factors and ergonomics, usability, accessibility, human interaction-
computer and the research of the user persons;

b) users and other stakeholders (or those who may represent their viewpoints);

c) experience in the field of application, subject matter experience;

d) marketing, brands, sales, technical support and maintenance, health and safety;

e) user interface, visual and product design;

f) technical writing, training, and user support;

g) user management, service management, and corporate governance;

h) business analysis, systems analysis;

i) systems engineering, hardware and software engineering, programming,
production/manufacturing and maintenance;

j) human resources, sustainability, and other stakeholders.

Projects benefit from the creativity and additional ideas of interaction and the
collaboration of team members who collectively have a wide base of
skills. An additional advantage of a multidisciplinary and multiperspective approach is that the
team members are more aware of the limitations and realities of the other disciplines;
for example, technical experts may be more sensitive to people's issues
users and users can be more aware of technical limitations.

6PLANNING OF PERSON-CENTERED DESIGN

6.1 Generalities

Person-centered design should be planned and integrated at all stages of the lifecycle.
product, that is, conception, analysis, design, application, testing, and maintenance.

6.2 Responsibility

The project planning managers must consider the relative importance of the
human factors/ergonomics in the project through an assessment:

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a) how usability relates to the purpose and use of the product, system, or service (by example, size, number of users, relationship with other systems, security issues or health, accessibility, specialized applications, extreme environments;

b) the levels of the different types of risk that can result from poor usability (by example, financial, scarce product differentiation, security, required level of usability, acceptance;

c) la naturaleza del entorno de desarrollo (por ejemplo, tamaño del proyecto, plazo de marketing, range of technologies, internal or external project, type of contract.

Note 1. The underestimation of the scope of user interaction is a common characteristic of projects that do not adequately plan person-centered design, like a system that Initially it was intended to be fully automated but ends up requiring interaction. significant for the user.

In general, the goal is to select the most appropriate techniques and procedures for identification. and mitigate the risks for person-system interaction.

Note 2. The descriptions of the methods for carrying out user-centered design activities are found in the standard ISO/TR 16982. The details of the person-centered design processes that They can be used to implement the guide in this document, which can be found in ISO 9241-220. ISO 9241-220 uses the ISO standard format for process models and also contains processes to ensure the content of person-centered design in the systems strategy and the introduction and operation of interactive systems. The details of the processes used by a company to define and address the range broader issues of products and processes raised by the person-centered approach can to be found in ISO/TS 18152. The IEC 62508 standard provides more guidance on a user-centered approach. person for systems where reliability is essential.

6.3 Contenido del plan

The planning of person-centered design must include:

a) determine the appropriate methods and resources for the activities described in the Chapter 7

b) define procedures to integrate these activities and their results with other activities of system development;

c) identify the individuals and the responsible organization(s) for the activities of person-centered design and the range of skills and perspectives they provide;

d) develop effective procedures to establish feedback and communication about person-centered design activities as they affect others design activities and "compensations," as well as methods to document them results of these activities;

e) agree on appropriate milestones for the person-centered activities to be integrated into the general process of design and development;

f) to agree on appropriate deadlines that allow for iteration, the use of feedback and the possible design changes to be incorporated into the project calendar.

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6.4 Integration with the project plan

The person-centered design plan must be part of the overall project plan. To ensure that the person-centered design plan is carried out and implemented properly effective, it should be subject to the same project disciplines (for example, responsibilities, change control) that other key activities. The aspects of the project plan related to person-centered design should be examined and reviewed properly as needs change throughout the life of the project.

6.5 Calendar and resources

In project planning, time and resources must be allocated to the focused activities. in the person. This should include time for iteration and incorporating feedback of the users, and to evaluate whether the design solution meets the requirements of the user personas.

More time should also be allocated for communication among team participants. design and the reconciliation of potential conflicts and balances that involve related issues with person-system interaction. The communication and additional debate to identify and resolve usability problems in the early stages of the project will allow for significant savings in later stages, when the changes are, inevitably, more costly.

Human-centered design activities should begin in the earliest phase of project (for example, as part of the process of formulating the initial concept of the product or The person-centered design continues throughout the entire life of the project.

7 PERSON-CENTERED DESIGN ACTIVITIES

7.1 Generalities

Once the need to develop a system, product, or service has been identified, and it having made the decision to use person-centered development, four must be considered user-centered design activities during the design of any interactive system:

a) understand and specify the context of use (see section 7.2);

b) the specification of the user's requirements (see section 7.3);

c) the production of design solutions (see section 7.4);

d) evaluate the design (see section 7.5).

Each of the person-centered design activities can use existing information. and produce new information that should be captured appropriately for the approach of chosen development.

These activities take into account the challenges listed below.

There are often a series of user groups and other stakeholders whose needs should be taken into account.

The context of use can be very diverse and can vary from one group of users to another. among different tasks.

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At the beginning of a project, it is unlikely that the requirements that can be captured are exhaustive.

Some requirements only arise once a proposed solution is available.

The needs of users can be diverse and potentially contradictory to each other and with those of other stakeholders.

Initial design solutions rarely meet all the needs of the person user.

It is difficult to ensure that all parts of the system are considered in an integrated manner.

At a high level, the activities of the person-centered design project correspond to the general stages of design and development, from requirements, through design, to the verification and validation. But at a more detailed level, these activities can be applied to obtain feedback on the initial design concepts before the requirements are finalized. The evaluation of approximate prototypes and design mockups potentials will help gain a deeper understanding of the person's needs user, as well as to provide initial information on design concepts. These activities they can also be applied during the reviews of an interactive system and can be useful for evaluate the systems in routine operation.

Person-centered design activities should be incorporated into the design approach. from the organization for the projects.

Person-centered design can start at any point (see section 7.2 'Understanding') and specify the context of use" or 7.3 "Specify the requirements of the user" or 7.4 Produce design solutions or 7.5 Evaluate the design depending on the project.

EXAMPLE 1 When the users and their tasks are well understood and documented, the The project begins by specifying the known requirements of the users.

EXAMPLE 2 When users complain about an existing system, the project begins. with a systematic evaluation of the system.

EXAMPLE 3 In a project to identify innovations in a system that is already successful, the the project begins by expanding the knowledge of the context of use.

EXAMPLE 4 When the user base of a product should be expanded to include the elderly people, the project begins with new design solutions based on a guide of design for the relevant age groups.

Person-centered design activities can be incorporated into design approaches. as diverse as object-oriented, waterfall, HFI (human factors integration), development agile and fast, among others.

Note. In agile development, the context of use of the information and the objectives for the design results person-centered is established at the beginning of a project. During each sprint, all are carried out the activities of person-centered design and are documented as necessary for communication within the project team.

Figure 1 illustrates the interdependence of person-centered design activities.

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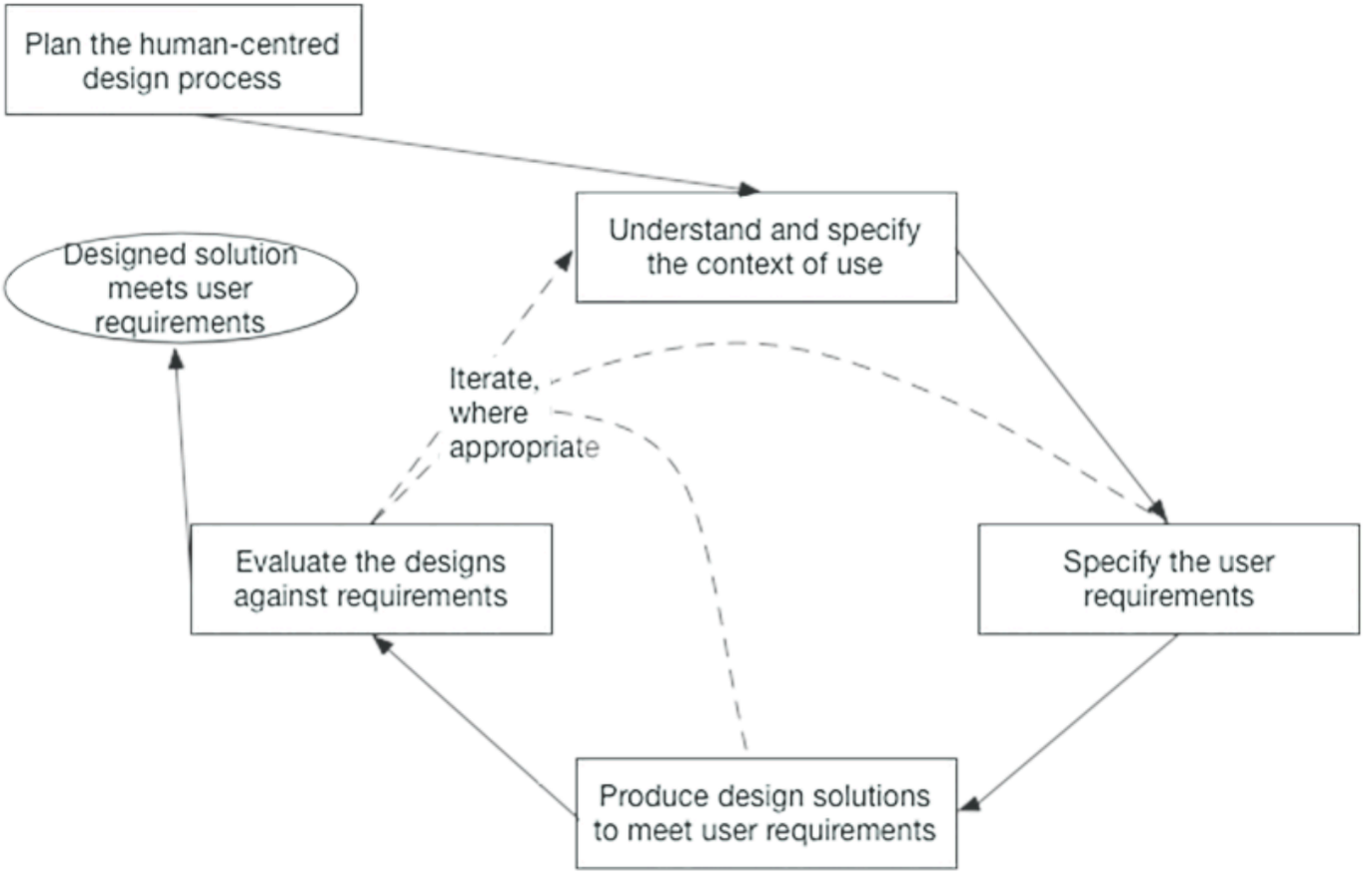


Figure 1 Interdependence of person-centered design activities.

7.2 Understanding and specification of the context of use

7.2.1 Generalities

The characteristics of users, tasks, and the organizational, technical, and physical environment. define the context in which the system is used. It is useful to collect and analyze information about the current context to understand, and then specify, the context that will apply to the future system. The analysis of existing or similar systems (including manual systems, if applicable) can, if still valid, provide information on a wide range of issues of context, including deficiencies and performance and satisfaction reference levels. It can reveal needs, problems, and limitations that, otherwise, might be overlooked, but that the future system should satisfy. In addition, some aspects of the current context will persist, even if the system is very novel. If an existing system is to be updated or improved, part this information will be available. If there are extensive results from the feedback from the user personas, help service reports and other data, these can provide a basis to prioritize modifications and changes in the system.

Note 1. A description of the use context can be a description of the current context of use or a description of the context intended for the design.

Nota 2.La norma ISO/TR 16982 proporciona información sobre una variedad de métodos que pueden utilizarse to gather and communicate this information.

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7.2.2 Description of the usage context

The description of the context of use must include the following

a) Users and other stakeholders: there may be a number of interest groups different users, as well as other stakeholders whose needs are important. Relevant groups must be identified and their relationship with development will be described. proposed in terms of key objectives and limitations.

b) The characteristics of users or groups of users: they must identify the relevant characteristics of the users. These may include ["knowledge","skill","experience","education","training","physical attributes","habits"] preferences and capabilities. If necessary, the characteristics of the different types of users, for example, with different levels of experience or ability physics. To achieve accessibility, products, systems, and services should be designed to be used by people with the widest range of abilities in the user populations they are intended for. This is a legal requirement in many countries.

Note. The ISO/IEC 29138-1 standard identifies a series of needs of users that should be to take into account to provide accessibility for people with disabilities.

c) The objectives and tasks of the users: the objectives must be identified. user personas and the general objectives of the system. They should be described characteristics of tasks that may influence usability and accessibility, for For example, the way in which users usually carry out tasks, the frequency and duration of performance, the interdependencies and the activities that should be carried out cable in parallel. If there are possible adverse consequences for health and safety (for example, an excessive workload caused by an inadequate work pace in a call center) or if there is a risk that the task will be performed incorrectly For example, making an incorrect purchase should also be identified. The tasks do not they should be described only in terms of the functions or characteristics provided by a product or system.

d) The environment or environments of the system: the technical environment should be identified, including the physical support, software programs, and materials. Additionally, the must describe the relevant characteristics of the physical, social, and cultural environment. Physical attributes include issues such as thermal conditions, lighting, spatial arrangement and the furniture. The social and cultural aspects of the environment include factors such as the work practices, the organizational structure, and attitudes.

7.2.3 Sufficient details to support the design

The usage context of the system should be described in sufficient detail to support the requirements, the design and the evaluation activities.

Note. The description of the context of use is a working document that is first prepared in terms general and then it is reviewed, maintained, expanded, and updated during the design and development process. For example, At an early stage of development, it may only be possible to identify the objectives of the tasks instead of the detailed activities of the tasks. It can also identify important design implications that they arise during the analysis.

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7.2.4 Specified usage context for design

The specified usage context for the design (that is, the context in which the system will be used) should be indicated in the user requirements specification to identify
Clearly the conditions under which the requirements apply. See INTE/ISO 9241-11 for more. information about the context of use and a sample report, and ISO 20282-1 for more information about the context of use for everyday products.

7.3 User Requirements Specification

7.3.1 Generalities

In most design projects, identifying people's needs
users and the specification of functional and other requirements for the product or system is an important activity. In the case of human-centered design, this activity must be expand to create an explicit statement of the user's needs in relation to
with the intended use context and the business objectives of the system.

Depending on the scope of the system, the user requirements may include changes. organizational and work styles reviewed and can suggest opportunities to combine products and services. If it is known that the proposed interactive system will affect organizational practice, the The development process should involve the stakeholders of the organization, with the aim to optimize both organizational and technical systems.

7.3.2 Identification of the needs of users and other stakeholders

The needs of users and other stakeholders should be identified.
taking into account the context of use. These should include what users need
achieve (instead of how to achieve it) and any restrictions imposed by the context of use.

7.3.3 Derivation of user requirements

The specification of the user's requirements must include:

a) the intended context of use;

b) requirements derived from the needs of the user and the context of use - by for example, it may be necessary for a product to be used outdoors;

c) the requirements derived from relevant knowledge, standards, and guidelines pertaining to ergonomics and user interface (for example, accessibility requirements are listed in the ISO 9241-20 and INTE/ISO 9241-171 standards;

d) the usability requirements and objectives, including measurable performance criteria usability and satisfaction in specific usage contexts; for example, one objective may be that 90% of the intended users can successfully divert an incoming call the voicemail, or the aesthetic design of a webpage to achieve a certain score of satisfaction for the user person;

e) the requirements derived from the organization's requirements that directly affect the user - for example, a call center system may require that calls from the customers are responded to within a specific timeframe.

The requirements of the users provide the basis for the design and evaluation of interactive systems to meet the needs of users.

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The requirements of the users are developed jointly with the specification of the general requirements of an interactive system and are part of it.

7.3.4 Resolution of commitments between the needs of users

Possible conflicts between the user's requirements should be resolved, for example, between precision and speed.

The justifications, the factors, and the weighting of the issues related to the human system for its use in compensations should be documented in a way that they can be understood in the future.

Note. Making these adjustments may require reviewing the initial assumptions and participation from the stakeholders.

7.3.5 Ensure the quality of the user requirements specifications.

The specification of the user's requirements should be:

a) expressed in terms that allow for subsequent testing;

b) verified by the relevant stakeholders;

c) internally consistent; and

It is updated as necessary during the life of the project.

7.4 Production of design solutions

7.4.1 Generalities

Design decisions have a significant impact on the user experience. The design person-centered aims to achieve a good user experience by considering it throughout the entire design process (see section 5.6).

Possible design solutions are developed based on the description of the context of use, the baseline evaluation results, the current state of the art in the field of applications, the guidelines and standards of design and usability, and the experience and knowledge from the multidisciplinary design team. As the possible solutions are detailed and evaluated Design, other requirements from users may arise.

The production of design solutions should include the following subactivities:

a) design the user tasks, the user-system interaction, and the user interface to meet the requirements of the user, taking into account all the experience of the user

b) specify the design solutions (for example, using scenarios, simulations, prototypes or mock-ups);

c) modify design solutions in response to person-centered evaluation user and feedback (see section 7.5 for more details on the evaluation of the user persona;

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d) communicate the design solutions to those responsible for their implementation.

7.4.2 Design of user tasks, user-system interaction, and user interface for satisfy the user's requirements, taking into account the entire experience of the user persona

7.4.2.1 Design Principles

Designing for the user experience is an innovative process that takes into account the satisfaction of the user (including emotional and aesthetic aspects), as well as the effectiveness and efficiency. Design involves a variety of creative approaches to achieve a good user experience.

The following principles (taken from the INTE/ISO 9241-110 standard) should be considered when design interactive systems:

suitability for the task;

b) the self-description;

c) compliance with the user's expectations;

d) suitability for learning;

e) controllability;

f) error tolerance;

g) suitability for individualization.

"Self-description" means that it is obvious to users in what dialogue they are engaged. where they are found within the dialogue, what actions can be taken and how they can be carried out.

Note 2. There are other design principles in other interaction standards between person and system, that They provide guidance to support this design activity. These are listed in the Bibliography.

7.4.2.2 Design of tasks and interaction between the user and the system

The proper design of the user-system interaction is based on a clear understanding of the context. of intended use, including the roles and tasks of the users and their results. This understanding allows for an appropriate allocation of functions, that is, the division of tasks of system between those carried out by human beings and those carried out by technology.

When the system is being developed for use within a specific organization, by example, a new system of bank branches, this activity may also involve design of work and organization (the INTE/ISO 9241-2 standard provides guidance on design of work and tasks).

Interaction design involves deciding how users will perform tasks with the system instead of describing what the system is like. For example, decisions at this point may include issues such as the choice of modality (for example, auditory, visual, and tactile) and the choice of the media (for example, text versus graphics, dialogue boxes versus assistants, mechanical controls versus electronic controls.

The interaction design should include:

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make high-level decisions (for example, initial design concept, essential outcomes);

- identify tasks and subtasks;

assign tasks and subtasks to the user and to other parts of the system;

identify the interaction objects necessary for the completion of tasks;

- identify and select the appropriate dialogue techniques (see the INTE/ISO 9241 standards - 112, ISO 9241-125, INTE/ISO 9241-129, ISO 9241-143, ISO 9241-154 and ISO 9241-161);

design the sequence and timing (dynamics) of the interaction;

design the information architecture of the user interface of an interactive system for allow efficient access to interaction objects.

Note. The order in which these activities are carried out depends on the type of interaction being designed. and it is in itself an iterative activity.

7.4.2.3 User Interface Design

For the detailed design of the user interface, there is a substantial body of knowledge of ergonomics and user interface, standards and guidelines that should be used to inform the design both hardware and software. These include standards from the ISO 9241 series on screens, input devices, dialogue principles, menus, information presentation, guide user and other user interface and accessibility guidelines. Many organizations also they have internal style guides for the user interface, product knowledge, and knowledge of users and other aspects of the context of use, such as the person's expectations user (see ISO standard 1503) and stereotypes. See the relevant standards of the ISO 9241 series. in Annex A.

7.4.3 Concrete design solutions

The use of scenarios, simulations, models, and mock-ups or other forms of prototypes allows designers communicate the proposed design to users and other stakeholders to obtain feedback.

The benefits include:

a) make the design proposals more explicit (this allows team members to design to communicate with each other and with users at the beginning of the development process;

b) allow designers to explore various design concepts before deciding on one;

c) allow the incorporation of user comments into the design in the first phases of the development process;

d) allow the evaluation of various iterations of a drawing or model and alternative designs;

e) improve the quality and completeness of the functional design specifications.

Simple prototypes are valuable in an early stage to explore design solutions. alternativas. Aunque puede haber un beneficio sustancial en hacer que las soluciones de diseño be as realistic as possible, the level of detail and realism should be appropriate for the topics that they need to be investigated. Investing too much time or money in the production of a prototype of Detailed work can lead to reluctance to change the design.

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7.4.4 Modification of design solutions based on evaluation centered on user persona and feedback

The feedback from the evaluation should be used to improve and refine the system (see section 7.5 for details on user-centered evaluation.

Note 1. Feedback reveals the strengths and weaknesses of the design solution and can provide new information about the user's needs and suggest areas where improvements can be made design.

The costs and benefits of the proposed changes should be evaluated and used to inform the decisions about what will be modified.

Note 2. The redesign effort depends on the nature of the problem; it can be small or it may require substantial resources, and the decision to redesign it is evaluated based on the criticality of the problem.

It is likely that the proposed changes based on an early assessment are the most profitable.

The project plans should allow enough time to make changes as a result of that feedback.

7.4.5 Communication of the design solution to those responsible for implementation

There are a variety of ways to communicate the design solution to teams and individuals. responsible for implementation or manufacturing. Effective means of communication can vary from providing the appropriate documentation, to producing revised prototypes, to including experts in person-centered design in the design and development team.

Cualquiera que sea la naturaleza del proyecto en su conjunto, debería haber un canal de sustained communication between those responsible for person-centered design and other members of the project team. When design solutions are communicated, they should be accompanied provide an explanation and justification of design decisions, especially when necessary make concessions.

Communication should take into account the limitations imposed by the project and the knowledge and understanding of the project team about ergonomics and design of the user interface.

7.5 Design Evaluation

7.5.1 Generalities

User-centered evaluation (evaluation based on the perspective of individuals)
User experience is a necessary activity in person-centered design.

Even in the early stages of the project, design concepts should be evaluated for gain a better understanding of the needs of the users. The actual use of a the product, system, or service is complex and, although the ergonomic orientation of the design can providing useful support to designers, user-centered evaluation is a essential element of person-centered design. However, the evaluation by the user personas (user-centered testing, see section 7.5.4) is not always not practical or profitable in all phases of the project.

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Under such circumstances, design solutions should also be evaluated in other ways.
- por ejemplo, usando modelos de tareas y simulaciones. Estos métodos siguen centrados en la the way in which users will experience the system, even though the users themselves do not participate directly. The user-centered evaluation can be used for:

a) gather new information about the needs of users;

b) provide information about the strengths and weaknesses of the design solution from la perspectiva de la persona usuaria (con el fin de mejorar el diseño);

c) evaluate if the requirements of the users have been met (which may include the assessment of conformity with international, national, local standards, business or legal); and

d) establish baselines or make comparisons between designs.

7.5.2 User-Centered Evaluation

The user-centered evaluation should include:

a) the allocation of resources both to obtain early feedback in order to improve the product as, in a later phase, to determine if the objectives have been met requirements;

b) plan the user-centered evaluation to align with the programming of the project;

c) the conduct of sufficiently thorough tests to provide results significant for the system as a whole;

d) analyze the results, prioritize the problems and propose solutions;

e) adequately communicate the solutions so that the design team can use them effectively.

7.5.3 User-centered evaluation methods

There are a variety of user-centered evaluation methods that can be used. to evaluate the designs. ISO/TR 16982 provides guidance on these and others Usability methods, as well as on the selection of the method or set of methods most appropriate.

Note. More information can be found in the rules listed in Annex A and in the Bibliography. recommendations and tests, checklists and other means of compliance with the criteria ergonomic.

To obtain valid results, the assessment should be carried out by evaluators. experienced, and should use appropriate methods.

User-centered evaluation is useful at all stages of the project, from the initial concept of design to its long-term use, which can provide information for future versions of the product, system, or service (see section 7.5.6). In the early stages of development and design, changes are relatively cheap. The more time is allowed for the process advances, and the more the system is defined, the higher the cost of modifications.

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Resources for evaluation should be allocated both to obtain feedback.
early with which to improve the product as, at a later stage, to validate that they have
satisfies the requirements of the users. The scope of this last evaluation (summative)
it should depend on the risks associated with non-compliance with the requirements.

Two widely used approaches for user-centered evaluation are:

- tests based on the user; and

evaluation based on inspection using guidelines or usability requirements and accessibility.

Note. Compliance with some software guidelines and standards can be checked automatically, which that can be useful for identifying basic problems. For example, some aspects of accessibility of Software can be evaluated using automated testing tools.

7.5.4 User-based Testing

User-based testing can be conducted at any stage of the design.

In a very early phase, user models, scenarios, or ...
sketches of the design concepts and asking them to evaluate them in relation to a real context. For example, a new check-out concept can be evaluated using a three-dimensional model, or simple screen sketches can be used to evaluate a new navigation design for mobile phones. These early tests can provide valuable information about the acceptability of the proposed design. The detailed aspects of the design can often be evaluated quickly and economically, for example, using paper versions of the dialogues proposed. A mockup of the interaction through simulated tasks is always necessary. real and in an appropriate context.

When prototypes are tested, users should perform tasks using the prototype instead of just watching demonstrations or a preview of the design. The information collected is used to drive design.

In a later phase of development, user-based testing can be carried out.
to evaluate whether the usability objectives have been met, including measurable criteria of usability and satisfaction performance, in the context or contexts of intended use.

A user-based testing method is field validation, that is, testing
of the designs or design concepts in real environments. In the case of software products, These tests are often referred to as 'beta' tests, in which a version is made available. initial software for its use, and users are informed that the product is not definitive and is still in the process of refinement. Hardware products can be manufacture in small quantities to conduct similar tests in the real world. The products fully developed can also be evaluated in the field to provide information for future developments.

The techniques that can be used to collect field validation data include data from the servicio de asistencia, reportes de campo, análisis de incidentes, reportes de cuasi accidentes, log files, defect reports, feedback from real users, data from performance, satisfaction surveys, health impact reports, design improvements, user observation and change requests.

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7.5.5 Inspection-based evaluation

Inspection-based evaluation can be valuable and cost-effective and can also complement user testing. It can be used to eliminate major issues before testing of the user and, therefore, make user testing more cost-effective.

The inspection-based evaluation is ideally conducted by usability experts who base your judgment on the previous experience of the problems encountered by the users and in their own knowledge of ergonomic guidelines and standards. The evaluation of several experts it can be combined to reduce individual bias. The inspection may involve the evaluator put yourself in the role of the user who works with the system, product, or service. Inspection-based evaluation can be supported by checklists, lists of user requirements, general guidance on usability, industry best practices, heuristics on usability, guidelines or standards. However, the effectiveness of the inspection always depends on the skills, experience, and knowledge of the evaluators.

Inspection-based evaluation is simpler and faster than user testing and, in In principle, it can take into account a wider range of users and tasks than the evaluation. based on the user (for example, to determine if a product meets the requirements of the user in contexts of use not selected for user testing). The inspection does not always find the same problems that would be found in based testing on the user. The inspection tends to emphasize obvious problems and may not be suitable for complex or novel interfaces. The greater the difference between knowledge and the the experience of inspectors and actual users, the less reliable the results will be. When appropriate, inspection-based evaluation can be conducted jointly. with domain experts of the application.

The relevant guidelines and standards are an important contribution to the design (see section 7.4.2), and compliance can be evaluated through inspection. Although it can be time-consuming. and requiring many resources, it may be necessary to check compliance, for example, with the web accessibility guidelines.

7.5.6 Long-term follow-up

A person-centered design process should also include long-term follow-up. from the use of the product, system, or service. This implies gathering contributions from people. users in different ways over a period of time.

The follow-up evaluation is often a formal part of the system evaluation and is carried out to take place within a specific timeframe, for example, six months to a year after la instalación del sistema. La evaluación de seguimiento a menudo pone a prueba el rendimiento del system and also collects data to determine if the needs and requirements of the people Users have been satisfied and if they are correctly established.

There is an important difference between short-term evaluation and long-term monitoring. Some effects of working with an interactive product, system, or service are not recognizable until that has been used over a period of time. Similarly, there may be resulting effects of external factors, such as unforeseen changes in legislation. It is not necessary to address these immediate problems, although the information obtained can still be used for the future modification or development of the product, system, or service.

Long-term performance data and reports on any health effects may provide valuable information. The criteria and measurements should be sufficiently sensitive to identify system faults, or system problems, as soon as possible.

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Note. Identifying unsafe behaviors is clearly preferable to recording accidents, and identifying the Mental or physiological overload is preferable to the recording of medical disorders.

SUSTAINABILITY AND HUMAN-CENTERED DESIGN

In modern society, a key issue is to promote socially responsible designs that consider sustainability. In terms of standardization, this means integrating and balancing the economic, social and environmental considerations.

ISO has committed to developing "standards for a sustainable world." The 1987 report of the United Nations Brundtland Commission, Our Common Future, defined sustainable development as satisfy the needs of the present without compromising the ability of future generations to satisfy their own needs.

Person-centered design directly supports the first two pillars of sustainability:

a)economic: adapt a design to the needs and capabilities of the users improves its utilization, quality, and efficiency, thus providing cost-effective solutions and reducing the probability that the products and services of the systems are wasteful or rejected by their users;

b)social: the adoption of a person-centered approach results in systems, products and services that are better for health, well-being, and engagement of its users, including users with disabilities.

Person-centered design also supports the environmental component by promotion of a design approach based on the complete life cycle. Explicitly encourages everyone involved in the design to consider the long-term implications of their system for its users and, therefore, for the environment. It is more likely to be maintained and applied continuously an approach that leads to usable products.

CONFORMITY

Compliance with this document is achieved through:

a) meet all requirements;

b) identify the applicable recommendations;

c) explain why certain recommendations are not applicable;

d) indicate whether the applicable recommendations have been followed or not.

If it is stated that a product or system meets the requirements and if it is considered that Following the applicable recommendations, the procedure used should be specified for determine how they have been fulfilled/followed. The details of the procedure are subject to negotiation. among the parties involved.

Annex B provides a means to both record the applicability of the recommendations as to inform that the requirements and recommendations have been met or followed applicable.

The users of this document can use the procedure and the forms of the Annex B or develop another procedure adapted to your particular needs.

Note. The ISO 9241-220 standard provides an evaluation model to demonstrate design capability. person-centered within a project or an organization.

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ANNEX A (INFORMATIONAL)

Summary of the ISO 9241 series

Table A.1 provides an overview of the structure of the ISO 9241 series.

The structure reflects the numbering of the original standard ISO 9241; for example, the screens were originally ISO 9241-3 and now they are from the ISO 9241-300 series. In each section, the "hundred" is a introduction to the section; for example, ISO/TR 9241-100 provides an introduction to the parts of software ergonomics.

Table A.1—Structure of ISO 9241 - Ergonomics of Human-Computer Interaction

Part	Title
1	Introduction
2	Guidance on task requirements
11	Usability: Definitions and concepts
20	Accessibility guidelines for information technology equipment and services and communication (ICT)
21–99	To assign
100	Software ergonomics
200	Person-system interaction process
300	Screens and hardware related to screens
400	Input physical devices - Principles of ergonomics
500	Ergonomics in the workplace
600	Environmental economics
700	Special application areas
800	To be assigned
900	Tactile and haptic interactions

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ANNEX B (INFORMATIVE)

Sampling procedure to assess applicability and conformity

B.1 Generalities

This appendix provides an example of a checklist (see Table B.1) that can be used to determine if the requirements of this document have been met and if they have been followed applicable recommendations.

This checklist contains all the requirements and recommendations of this document, presented in sequence, but cannot be used in isolation from the complete content of this part of the document.

It is worth noting that the described procedure is provided for guidance purposes and is not a process exhaustive that should be used as a substitute for the own standard.

The use of the checklist provides a foundation for:

determine which of the recommendations are applicable;

determine if the requirements have been met and if the recommendations have been followed applicable; and

- provide a list in support of a declaration of conformity that demonstrates that it has fulfilled all requirements, and a systematic list of all applicable recommendations that have been followed.

Several of the requirements and recommendations in this document have more than one component. These They are presented in lists. The satisfaction of the requirement or recommendation depends on the consideration. of each component and not of the requirement or recommendation itself. Therefore, each element of these lists are presented on a line separate from the checklist and the line that contains the the requirement is marked with a gray fill. The completed checklist can be used in support for the statements regarding the compliance of a project with this document.

Provide a list of all the requirements and recommendations for which the achievement has been made. conformity.

B.2 How to use the checklist

The chapter and section numbers are presented in the first column of the table, with the title corresponding or the requirements/recommendations in the second column. The third column is used to indicate whether the requirement or recommendation of each chapter or section is applicable or not. All the requirements already have S (for "yes") inserted in column 3. All the other chapters and Sections should be checked in relation to the context of the project, and Y or N (for 'no') They should be indicated in column 3, as appropriate.

For each recommendation, in the relevant chapter or section of this document, there provide information about the applicable circumstances. If the recommendation is not applicable, it will be indicated in column 3 of the 'Applicability' section of the table, and a brief explanation will be provided in column 4 "Non-applicable reasons".

Checking whether a requirement or recommendation has been met involves examining all the elementos que se ha demostrado que son aplicables en la columna 3 y determinar si el proyecto que It is being evaluated whether it meets those requirements and recommendations (when applicable). The method Exactly, to make this determination can vary.

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The 'Compliance' section of the checklist provides space in columns 5 and 6.
to indicate the decision of whether each applicable requirement or recommendation has been met (Y) or not (N).

Every chapter or section that is not fulfilled should be accompanied by a brief note in the column.
7 in which the reasons for it are explained. Column 7 can also be used to record
information about the method used.

B.3 Copy the checklist

The users of this document can freely reproduce the table contained in this.
annex to demonstrate your compliance with this document.
The versions editables of the list of verification they are available in https://standards.iso.org/iso/9241/210/ed-2/en.

Table B.1 - Checklist for assessing applicability and compliance with this document

Chapter/ section	Requirements or recommendations	Applicability		Conformity		
		Si/ No	Motivo no aplicable	Si	No	Comentarios
5	**Principles of user-centered design**					
5.1	Whatever the process of design y the assignment of responsibilities and roles adapting a person-centered approach should to follow the principles enumerated [in section 5.1].					
5.2	Products, systems, and services they should be designed taking into account count the people that the they will utilize, as well as other groups interested parties, including those who can be affected (directly or indirectly) by its use.					
5.2	All should be identified. user groups and parts relevant stakeholders. [see also section 7.2.2 a)]					
5.3	The participation of the people users should be active.					
5.3	The involved users should to have capabilities characteristics and experience that reflect the range of users for the that the system is being designed. [See also section 7.2.2 b)]					
5.4	The evaluation centered on the the user persona should take place as part of the final acceptance of product to confirm that they have fulfilled the requirements.					

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Chapter/ section	Requirements or recommendations	Applicability		Conformity		
		Si/ Not	Motivo no applicable	Si	No	Comentarios
5.5	Iteration should be used for remove progressively the uncertainty during development of interactive systems.					
5.6	It should also be taken into account the user's experience with previous systems or others systems and issues such as the brand and advertising.					
5.6	The fortresses, limitations preferences and expectations of the user personas should be taken into account take into account when specified the activities carried out by the users and their functions that technology plays.					
5.6	In general, the users representatives should participate in decisions related to the assignment of functions.					
5.6	The resulting human activities from the assignment of functions they should form a set of tasks that are meaningful in their set for the user.					
5.7	Design teams focused on the person does not have to be big, but the team should be diverse enough as to collaborate in the design and the implementation of decisions of compensation at the moments adequate.					
6	**Person-centered design planning**					
6.1	It should be planned and integrated into all the phases of the life cycle of product the design centered on the person.	S				
6.2	The project planning managers must consider the importance relative to the human factors/ergonomics in the project through an evaluation:					
6.2 a)	how usability is related to the purpose and use of the product, system or service	S				
6.2 b)	the levels of the different types of risks that may result from a poor usability	S				
6.2 c)	the nature of the environment of development	S				
6.3	The planning of person-centered design must include:					
6.3 a)	determine the methods and resources suitable for the activities described in chapter 6	S				

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Chapter/ section	Requirements or recommendations	Applicability		Compliance		
		Si/ Not	Motivo no applicable	Si	No	Comentarios
6.3 b)	define procedures for integration these activities and their results with other development activities of the system	S				
6.3 c)	identify the people and the responsible organization(s) for user-centered design activities in the person and the range of skills and viewpoints that they provide	S				
6.3 d)	develop effective procedures to establish the feedback and the communication about the design-centered activities the person to the extent that affect other design activities and the 'compensations', as well as methods to document the results of these activities	S				
6.3 e)	set appropriate milestones for the actividades centradas en la persona that they integrate into the process general design and development	S				
6.3 f)	agree on suitable deadlines that allow the iteration, the use of the feedback and the possible design changes that are incorporate into the calendar of project	S				
6.4	The design-centered plan of the a person must be part of the plan general of the development project of the system.	S				
6.4	To ensure that the design plan person-centered care be followed it is implemented effectively, should be subject to the same project disciplines (for example, responsibilities, control of changes) like other activities key.					
6.4	They should to be examined y revised adequately the aspects of the project plan related to user-centered design in the person as they change the needs throughout life of the project.					
6.5	Time and resources must be assigned to the project planning to the person-centered activities.	S				

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Chapter/ section	Requirements or recommendations	Applicability		Conformity		
		Si/ Not	Motivo no applicable	Si	No	Comentarios
6.5	The plan must include time for the iteration and the incorporation of the feedback from the person user, and to evaluate if the solution design meets the requirements of the user	S				
6.5	More time should be dedicated to the communication among the participants from the design team and to reconcile the potential conflicts and the compensations that involve issues related to the person-system interaction.					
6.5	User-Centered Design Activities in person should start in the earliest phase of the project.					
6.5	The aspects of user-centered design the person of the project plan they should be reviewed throughout the entire life of the project.					
7	**Person-centered design activities**					
7.1	Existen cuatro actividades de diseño centradas en la persona que se deben llevar a head during the design of any interactive system:					
7.1 a)	Understanding and specification of usage context	S				
7.1 b)	Specification of needs of the user	S				
7.1 c)	Produce design solutions	S				
7.1 d)	Design Evaluation	S				
7.1	User-centered design activities should be incorporated into the person in the design approach of the organization for the projects.					
7.2.2	The description of the context of use it must include the following:	S				
7.2.2 a)	The groups must be identified relevant and should be described their relationship with the proposed development in terms of objectives y key limitations.	S				
7.2.2 b)	If must identify the pertinent characteristics of the user personas.	S				
7.2.2 b)	If necessary, they should be defined. the characteristics of the different types of users.					

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Chapter/ section	Requirements or recommendations	Applicability		Conformity		
		Si/ Not	Motivo no applicable	Si	No	Comentarios
7.2.2 b)	To achieve accessibility, the products, systems, and services they should be designed to be used by people with the most wide range of capabilities in the user populations to which they are destined.					
7.2.2 c)	The objectives must be identified the users and the objectives system general.	S				
7.2.2 c)	If must describe the characteristics of the tasks that can influence usability and the accessibility.	S				
7.2.2 c)	All should be identified possible adverse consequences for health and safety.					
7.2.2 c)	If there is a risk that the task may be completed incorrectly, should be identified.					
7.2.2 c)	Tasks should not be described. only in terms of the functions o characteristics provided by a product or system.					
7.2.2 d)	The technical environment must be identified. including the physical support, the computer programs y the materials.	S				
7.2.2 d)	The characteristics must be described relevant to the physical and social environment, organizational and cultural.	S				
7.2.3	It should be described with enough detail the context of use of system to support the requirements, the design and the activities of evaluation.	S				
7.2.4	It should be specified as part of the specification of the requirements of the user the context of use expected to clearly identify the conditions under which they apply the requirements.					

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Chapter/ section	Requirements or recommendations	Applicability		Conformity		
		Si/ Not	Motivo no applicable	Si	No	Comentarios
7.3.1	The identification of needs of the users and the specification of requirements functional and other types of product or system must be expanded to create an explicit statement of the requirements of the people users in relation to the context intended use and objectives businesses of the system.					
7.3.1	If it is known that the interactive system proposed will affect the practice organizational, the process of development should involve the parts interested of the organization in the design process with the aim of optimizing both the organizational systems such as the technicians.					
7.3.2	The needs of people users and other stakeholders they should be identified, having consider the context of use.					
7.3.2	The needs of people users and other stakeholders they should include what people users need to achieve (instead of on how to achieve it) and any restriction imposed by the context for use.					
7.3.3	The specification of the user's requirements must include:					
7.3.3 a)	the intended context of use	S				
7.3.3 b)	the needs derived from the needs of the user and of the context of use	S				
7.3.3 c)	the requirements derived from the relevant ergonomics and of the conocimientos, normas y directrices from the user interface	S				
7.3.3 d)	usability requirements and objectives, including measurable criteria of usability performance and of satisfaction in specific contexts for use	S				
7.3.3 e)	needs derived from the needs organizational what directly affect the user	S				
7.3.4	Possible issues should be resolved. conflicts between the needs of the users.					

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		Si/ Not	Motivo no applicable	Si	No	Comentarios
7.3.4	The justifications, the factors and the pondering of the issues related to interaction persona-system for its use in any compensation should to inform oneself in such a way that can be understood in the future.					
7.3.5	The specification of the user's requirements should be:					
7.3.5 a)	in terms that allow the conducting follow-up tests					
7.3.5 b)	verified by the stakeholders relevant					
7.3.5 c)	internally coherent					
7.3.5 d)	it will be updated as necessary during the life of the project					
7.4.1	The production of design solutions should include the following subactivities:					
7.4.1 a)	design user tasks, the user-system interaction and the user interface for meet the requirements of the user persona, taking into account the overall experience of the person user					
7.4.1 b)	concretize the design solutions					
7.4.1 c)	modify the design solutions in response to the evaluation focused on the user and to the feedback					
7.4.1 d)	communicate the design solutions a the responsible of on implementation					
7.4.2.1	The following principles (taken from the INTE/ISO 9241-110 standard) should take into account when designing interactive systems:					
7.4.2.1 a)	suitability for the task					
7.4.2.1 b)	self-description					
7.4.2.1 c)	compliance with the expectations of the user					
7.4.2.1 d)	suitability for learning					
7.4.2.1 e)	controllability					
7.4.2.1 f)	error tolerance					
7.4.2.1 g)	suitability for individualization					
7.4.2.2	The interaction design should include:					
7.4.2.2 a)	make high-level decisions					
7.4.2.2 b)	identify tasks and subtasks					
7.4.2.2 c)	assign tasks and subtasks to the user and to other parts of the system					
7.4.2.2 d)	identify the interaction objects necessary for the realization of the tasks					
7.4.2.2 e)	identify the dialogue techniques appropriate					

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		Si/ Not	Motivo no applicable	Si	No	Comentarios
7.4.2.2 f)	design the sequence and the time (dynamic) of the interaction					
7.4.2.2 g)	design the architecture of user interface information of an interactive system for allow efficient access to the interaction objects					
7.4.2.3	Ergonomics and knowledge of the user interface, the standards and guidelines should utilizarse para informar el diseño both hardware and software of the person's interface user.					
7.4.3	The level of detail and realism [of the prototypes should be appropriate for the topics that need to be investigated.					
7.4.4	The feedback of the evaluation should be used for improve and refine the system.					
7.4.4	The costs and benefits of proposed changes should be evaluated and used to inform the decisions about what is will modify.					
7.4.4	The project plans should provide enough time to carry out the changes as a result of that feedback.					
7.4.5	There should be a channel of sustained communication between the responsible for user-centered design the person and other members of the project team.					
7.4.5	When the solutions are communicated of design, they should be accompanied of an explanation and justification of the decisions of design especially when sea necessary to make concessions.					
7.4.5	The communication [of the details of design should take into account the limitations imposed by the project and the knowledge and the project team understanding about ergonomics and design of the user interface.					

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Chapter/ section	Requirements or recommendations	Applicability		Conformity		
		Si/ Not	Motivo no applicable	Si	No	Comentarios
7.5.1	The evaluation focused on the user persona (evaluation based from the person's perspective User activity is a necessary activity. in person-centered design.	S				
7.5.1	Even in the early stages of project, the design concepts they should be evaluated to obtain a better understanding of the needs of people users.					
7.5.1	If the tests based on the user persona is not practical or profitable at a particular stage of a project, the solutions of design should be evaluated from other perspectives ways.					
7.5.2	The user-centered evaluation should include:					
7.5.2 a)	assign resources both to obtain early feedback for improve the product, as for determine later if they have fulfilled the requirements					
7.5.2 b)	planning the evaluation focused on the user so that it adjust the schedule of the project					
7.5.2 c)	the carrying out of tests it sufficiently exhaustive as to provide meaningful results for the system as a whole					
7.5.2 d)	analyze the results, prioritize them problems and propose solutions					
7.5.2 e)	to communicate adequately the solutions for the team of design can use them effectively					
7.5.3	To obtain valid results, the evaluation should be carried out by experienced evaluators.					
7.5.3	To obtain valid results, the evaluation should use methods appropriate.					
7.5.3	The resources for the evaluation they should be assigned both obtain a feedback early with which to improve the product like, at a stage later, to validate if they have meets the requirements of the user personas.	for				

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		Si/ Not	Motivo no applicable	Si	No	Comentarios
7.5.3	The scope of this latest evaluation (summative) should depend on the magnitude of the associated risks with the non-compliance of the requirements.					
7.5.4	When the prototypes are tested, users should perform tasks using the prototype instead of just limiting oneself to seeing demonstrations or a preview of the design.					
7.5.6	A design process focused on person should include a seguimiento a largo plazo del uso del product, system or service.					
7.5.6	The criteria and measurements [for the long-term monitoring should be sensitive enough to identify failures of the system, or system problems, as soon as possible.					

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[3] In plans to be replaced by ISO 9241-220.

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